

Lab 10: Kuwahara Filter

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RGB2HSV produces brightness $V = \max(R, G, B)/255$. For radius 2, four 3×3 windows are compared by V variance; the RGB mean of the lowest-variance window becomes the output. The CPU and global-memory GPU compute the same windows. The shared GPU cooperatively loads a 20×20 RGB/V tile including halo for a 16×16 block, then synchronizes. The two-pixel border is preserved. The notebook checks both GPU outputs and reports CPU/global and global/shared speedups. RGB2HSV time is separate. Neighboring x threads access adjacent pixels; shared tiles reuse overlapping neighborhoods. No claim of eliminated bank conflicts is made without profiling.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.

mode	cpu_ms	gpu_ms	speedup_cpu	max_error	global_over_shared	rgb2hsv_ms
global	6720.26	0.88644	7581.17	0	0.970082	0.126044
shared	6720.26	0.913779	7354.36	0	0.970082	0.126044

